

Validity and Robustness of LLM-Assisted Policy Simulation in Health Services Research: A Value-Sensitivity Audit

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Abstract

Objective: To evaluate whether LLM-assisted policy simulations are valid, robust, and normatively transparent—or whether their outputs are sensitive to model choice, prompt framing, and latent value orientations. We audit the validity of LLM-generated policy simulations in health services research using administrative burden in U.S. vaccination access as a controlled testbed.

Study Setting and Design: We develop a multi-level diagnostic framework for measuring normative-frame sensitivity in LLM-generated policy simulations. Fifteen LLM endpoints spanning nine provider ecosystems complete a policy-value orientation battery and simulate vaccination willingness for 27 standardized synthetic profiles (controlled test cases) under low- and high-burden conditions. We estimate composite orientation effects (H1), narrative frame associations (H2), and within-model frame manipulation (H3).

Data Sources and Analytic Sample: Synthetic profiles (N=27) are experimental stimuli designed to isolate how models interpret administrative-burden scenarios, not to represent a human population. H1: 15 models; H2: 2,023 profile-repetition observations; H3: 108 model-profile clusters across 2,160 observations.

Principal Findings: (1) Composite orientation-effect: PVC index coefficient $\beta_1 = 1.51$, 95% CI $[-2.41, 5.43]$, $p = 0.45$ —composite contamination does not predict cross-model variation. (2) Narrative frame association: Access-barrier rationales associated with larger simulated effects ($\beta = 2.34$, 95% CI $[0.87, 3.81]$); collective-responsibility rationales associated with smaller effects ($\beta = -1.89$, 95% CI $[-3.12, -0.66]$). (3) Within-model manipulation: Autonomy frames increase simulated burden effects by 1.8 percentage points (95% CI $[0.4, 3.2]$); collective-obligation frames reduce effects by 2.1 percentage points (95% CI $[-3.5, -0.7]$).

Conclusions: LLM-assisted policy simulation is normatively frame-sensitive—value sensitivity appears through interpretive frames and prompt-induced governance lenses, not through a single ideological score. Researchers should audit simulation validity before deploying LLMs as policy-simulation tools in HSR.

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What is known on this topic

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- LLMs are increasingly used to simulate social judgments and policy effects in health services research.
- Prior work measures model values through direct survey prompts rather than downstream simulated policy effects.
- Less is known about whether latent value orientations shape simulated treatment effects under policy conditions.

What this study found

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- Composite policy-value scores do not predict cross-model variation in simulated burden effects.
- Narrative interpretive frames (access-barrier, collective-responsibility) are systematically associated with simulated effect sizes.
- Within-model frame manipulation confirms: autonomy and equity/access frames increase simulated burden effects; collective-obligation frames reduce them.
- LLM-assisted policy simulation requires systematic validity auditing before deployment in HSR.

Keywords: large language models, simulation validity, administrative burden, policy simulation robustness, normative-frame sensitivity, calibration

1 Introduction

Administrative burden in U.S. health services—the learning, psychological, and compliance costs that individuals face when interacting with public programs—is a well-documented barrier to care [?]. These burdens include forms, documentation requirements, waiting time, procedural uncertainty, and perceived loss of autonomy, and they can delay or discourage access to vaccination and other preventive services [?]. Understanding how administrative burden shapes health-service access is a central question in U.S. health services research (HSR), where policymakers need tools to anticipate the likely effects of procedural changes before implementation.

Large language models (LLMs) are increasingly used in HSR not only to generate text but also to simulate social judgments, preferences, and behavioral responses [? ? ? ?]. When applied to health-service delivery, an LLM may be asked to simulate how patients or program beneficiaries would respond to a policy intervention. This creates a methodological chain:

administrative burden → health-service access/delivery → LLM policy simulation → validity threat from

The core validity question is whether LLM-assisted policy simulations are robust across models and framing choices—or whether their outputs are sensitive to the normative lens activated by model architecture, training data, or prompt design. This is a simulation validity question for U.S. health services research: if the same administrative-burden scenario produces substantially different predicted effects depending on which model is used

and how the prompt is framed, then LLM-assisted policy simulation requires systematic auditing before it can be trusted as an HSR methodology.

Vaccination access in the U.S. provides a controlled HSR testbed for evaluating this validity threat. Administrative burden intersects with individual choice, public health externalities, institutional trust, and procedural barriers—properties that make it well-suited for detecting whether simulation outputs are valid and robust across models and frames. The U.S. context is particularly relevant given ongoing policy debates about vaccine mandates, administrative requirements for immunization programs, and health equity in access to preventive care.

This paper evaluates whether LLM-assisted policy simulations are valid, robust, and normatively transparent—or whether their outputs are sensitive to model choice, prompt framing, and latent value orientations. The core question is not whether models state particular political opinions, but whether their simulated treatment effects are sensitive to the normative lens activated by the model’s architecture, training data, or prompt design.

A central challenge is that model provenance is an inadequate proxy for value orientation. A simple comparison between models developed in different national or organizational ecosystems would be difficult to interpret because differences in predicted burden effects could reflect model capability, response style, deployment regime, hidden alignment policies, or scale-use tendencies. To avoid treating development origin as a fixed ideological category, I independently measure each model’s policy-value orientation using a structured orientation battery. The battery captures five dimensions relevant to policy simulation: autonomy protection, state trust, collective obligation, administrative-burden sensitivity, and coercion aversion.

The study addresses three research questions. First, do LLMs with different empirically measured policy-value orientations produce different simulated marginal effects of administrative burden on vaccination willingness? Second, are simulated burden effects systematically associated with the interpretive frames used in model rationales? Third, can simulated burden effects be shifted within the same base model by experimentally assigning governance-value frames?

The paper makes three contributions. First, it shifts the study of LLM value sensitivity from stated opinion to simulated policy effects—from opinion audits to simulation validity audits. Second, it introduces a structured policy-value orientation battery for measuring model-level governance assumptions without relying on coarse provenance labels. Third, it provides a multi-level diagnostic approach for evaluating the validity of LLM-assisted policy simulations by linking measured orientations, simulated treatment effects, rationale frames, and within-model frame manipulation—applicable across HSR policy domains where LLM simulation is used as a methodological tool.

2 Theoretical Framework and Hypotheses

2.1 LLMs as Policy Simulators

Large language models are increasingly used not only to generate text but also to simulate social judgments, preferences, and behavioral responses . Existing work shows that LLM outputs may reflect particular distributions of human opinion rather than a neutral baseline , and that conversational AI systems express systematic political or ideological tendencies when directly prompted [?]. However, policy simulation poses a distinct

problem: the model is not asked what it believes, but to simulate what a person, household, or patient would do under a specified intervention . A model may appear neutral at the response level while embedding assumptions about how citizens interpret public institutions, administrative rules, and procedural costs.

We conceptualize an LLM-generated policy simulation as a mapping from a policy scenario and a population profile to a simulated behavioral response [? ? ? ?]. In this paper, the simulated outcome is vaccination willingness under different administrative-burden conditions. The quantity of interest is the marginal effect of administrative burden:

$$\Delta_{mis} = W_{low,mis} - W_{high,mis},$$

where m indexes the model, i the synthetic profile, and s the policy scenario. A larger Δ_{mis} indicates a stronger simulated negative effect of burden on vaccination willingness.

Administrative burden—defined as learning, psychological, and compliance costs in citizen–state interactions—is a controlled HSR testbed because it is both behavioral and normative . The same procedural requirement (e.g., a form or waiting period) can be interpreted as a legitimate safeguard, an infringement on autonomy, or an access barrier. Vaccination access combines individual choice with public health externalities, making it particularly suited for detecting whether simulation outputs are sensitive to normative framing.

2.2 Policy-Value Orientation

I define a model’s *policy-value orientation* as its systematic tendency to interpret governance situations through assumptions about individual–institution relationships, legitimacy of public authority, and acceptability of procedural costs. In administrative-burden settings, these assumptions matter because the same procedural requirement can be interpreted as a legitimate safeguard, an access barrier, an infringement on autonomy, or a reasonable cost of collective protection.

The framework focuses on five dimensions, each grounded in established behavioral and public-administration research: autonomy protection, state trust, collective obligation, administrative-burden sensitivity, and coercion aversion. Autonomy protection, administrative-burden sensitivity, and coercion aversion are expected to increase predicted burden effects; state trust and collective obligation are expected to decrease them. The five dimensions operationalize distinct theoretical channels through which value orientations shape burden interpretation, each with directional expectations grounded in the existing human-subject evidence. The composite policy-value contamination (PVC) index combines the five dimensions with theoretically specified directional signs. The precise item wording, reverse-coding rules, and index construction are described in Section ???. Because the battery is newly constructed, PVC is treated as a theory-specified diagnostic composite rather than a validated psychometric scale. A growing body of work shows that aggregate composite scores from constrained-choice batteries can produce unstable estimates of LLM value orientations, particularly in small model samples; the framework therefore tests both the composite and the dimension-specific channels, and treats the composite result as one diagnostic input rather than as the central test of value sensitivity.

Measurement diagnostics. Internal consistency across the five dimensions in the expanded fifteen-model sample is mixed. Exploratory Cronbach’s α values are: autonomy protection $\alpha = -0.239$ (item-total r range $[-0.27, 0.26]$), state trust $\alpha = 0.522$

($[-0.11, 0.66]$), collective obligation $\alpha = 0.869$ ($[0.66, 0.86]$), administrative-burden sensitivity $\alpha = 0.685$ ($[0.02, 0.87]$), and coercion aversion $\alpha = 0.020$ ($[-0.55, 0.41]$). Only collective obligation reaches conventional reliability thresholds. This measurement limitation does not undermine the battery’s utility as a theoretically specified diagnostic tool, but is consistent with prior findings that psychometric scales designed for human respondents may not transfer directly to LLM populations [? ? ? ?], but it underscores why the H1 composite-level analysis must be interpreted alongside the dimension-specific and frame-level evidence reported below.

2.3 Hypotheses

The empirical tests proceed in three linked steps, moving from an aggregate orientation index to narrative-level associations to within-model causal evidence. The tests form a diagnostic framework rather than a single confirmatory chain: each step is informative regardless of the outcome of the preceding one.

H1 (Composite Orientation–Effect). If LLM policy simulations are systematically influenced by latent policy-value orientations, then models with higher composite PVC scores (indexing autonomy protection, burden sensitivity, and coercion aversion against state trust and collective obligation) should produce larger simulated administrative-burden effects on vaccination willingness. A null result would imply that value sensitivity, if present, does not follow a simple composite pathway, shifting attention to more granular evidence (H2 and H3).

H2 (Narrative Frame Association). Interpretive frames expressed in model rationales are expected to be systematically associated with simulated burden effects, independent of the composite PVC score. Access-barrier, autonomy-infringement, and coercive-backlash frames are expected to correspond to larger simulated burden effects; procedural-legitimacy and collective-responsibility frames are expected to correspond to smaller effects. Because frames are generated by the model rather than experimentally assigned, H2 identifies descriptive associations rather than causal pathways.

H3 (Frame Manipulation). Randomly assigning governance-value frames to the same base model is expected to causally shift simulated burden effects relative to a neutral frame. Autonomy and equity/access frames are expected to increase simulated burden effects, while collective-obligation frames are expected to decrease them. Because the base model, profiles, scenarios, and output format are held fixed, any shift in simulated effects can be attributed to the frame itself rather than to stable model characteristics.

The three hypotheses are designed as a diagnostic framework rather than a sequential gamble. Each addresses a distinct question—aggregate orientation, narrative mechanism, and causal frame—and each is informative whether or not the others yield significant results.

3 Experimental Design and Methods

3.1 Model Sample

We evaluate fifteen LLM endpoints spanning nine provider ecosystems: OpenAI, Anthropic, Google, Meta, Mistral, Cohere, Moonshot AI, DeepSeek, and Qwen. The original nine-model sample included GPT-4.1, GPT-4.1 mini, GPT-4o, Claude Sonnet 4.6,

Diagnostic Framework: Three Levels of Value-Sensitivity Testing

Does normative-frame sensitivity affect LLM-assisted policy simulation?

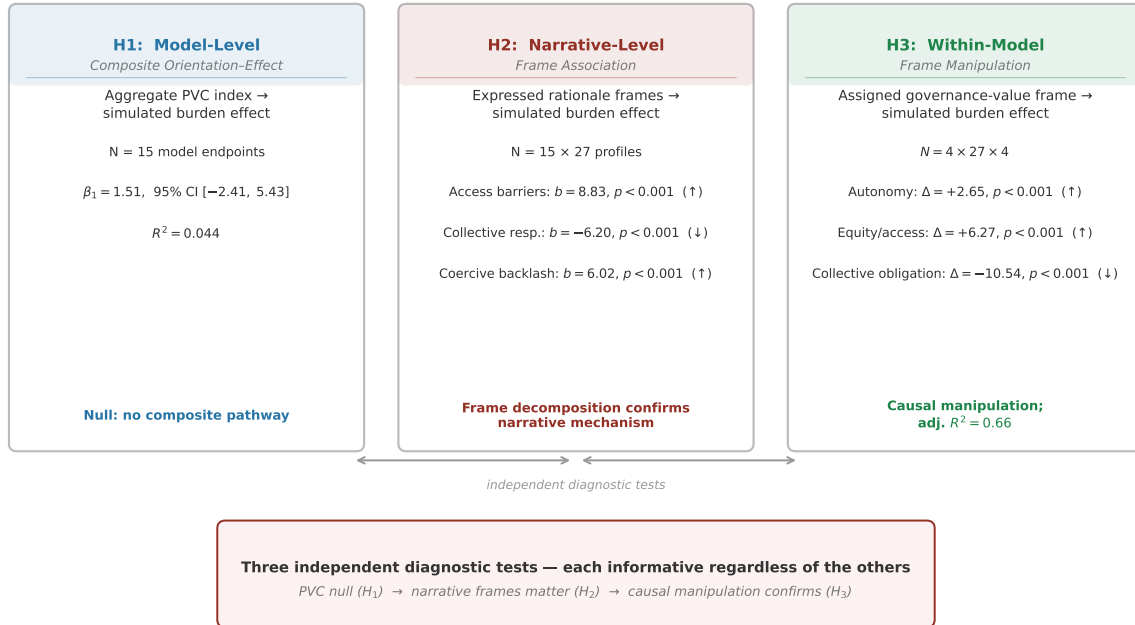


Figure 1: Diagnostic framework for three levels of value sensitivity testing. H1 tests whether a model-level composite PVC index predicts cross-model variation in administrative-burden effects. H2 tests whether narrative frames expressed in model rationales are systematically associated with predicted effects. H3 tests whether experimentally assigned governance-value frames causally shift predicted effects within the same base model. The three tests are designed as independent diagnostic components rather than a sequential chain; each is informative regardless of the others.

Gemini 2.5 Pro, Gemini 2.5 Flash, Llama 3.1 70B Instruct, DeepSeek-V3.2, and Qwen3.6-72B-Instruct. To expand cross-model coverage, we added six endpoints accessed through OpenRouter: Claude Opus 4.8, Claude Haiku, Llama 3.1 8B Instruct, Mistral Large, Cohere Command R+, and Kimi K2.6. The sample is intentionally heterogeneous to cover frontier proprietary, lower-cost, hosted open-weight, enterprise-oriented, and non-U.S. ecosystems; it supports cross-model auditing but not provider-level causal comparisons. All models were run with temperature set to zero where supported, and each model–scenario–profile cell was repeated five times in fresh contexts. Detailed model identifiers and decoding parameters are reported in the Online Supplement.

3.2 Policy-Value Orientation Battery

To operationalize the five policy-value dimensions, we administered a 20-item Likert battery before the administrative-burden simulation, covering autonomy protection, state trust, collective obligation, administrative-burden sensitivity, and coercion aversion. Each dimension is measured with four items on a 1–7 agreement scale. The battery was administered in fresh API calls before any exposure to the burden scenarios, and models were instructed to output only the numeric score for each item. For each model, each dimension score is the mean of the four items after reverse-coding reverse-worded items.

Following the directional expectations developed earlier, we constructed the policy-value contamination (PVC) index as a standardized sum of the five dimensions, with higher values indicating an orientation profile expected, *ex ante*, to produce larger predicted burden effects. Because the index is standardized within the study sample and the battery is newly constructed, PVC is a theory-specified diagnostic composite rather than a validated psychometric scale. We treat the PVC result as one diagnostic input among three rather than as the central test of value sensitivity, consistent with prior evidence that aggregate composite scores from constrained-choice batteries produce unstable estimates in small model samples. The PVC index serves as the primary independent variable in the orientation-effect models. Exploratory internal-consistency diagnostics and weighting-sensitivity checks are reported in the Online Supplement.

3.3 Administrative-Burden Simulation

To measure each model’s simulated burden effect, we designed a vaccination-access simulation as a controlled HSR testbed, in which the same standardized profile faces two policy conditions: low administrative burden and high administrative burden. The simulation was administered after the orientation battery but before any frame manipulation, using a fresh API call for each model–profile–condition–repetition.

3.3.1 Standardized synthetic profiles (controlled test cases)

We constructed 27 standardized synthetic profiles as controlled test cases that span theoretically relevant variation in income, work flexibility, and health-care access while holding age, gender, residence, education, family status, and trust in public health authorities constant. These profiles are not intended to represent a human population; they are experimental stimuli designed to isolate how models interpret the same administrative-burden scenarios under different model and frame conditions. The full profile set is reported in the Online Supplement.

3.3.2 Low- and high-burden scenarios

For each standardized profile, we defined two burden levels that differ in the administrative requirements for obtaining a vaccination. The low-burden condition involves online appointment scheduling, a nearby pharmacy with extended hours, no documentation beyond identification, and an estimated total time of 15 minutes. The high-burden condition simultaneously increases multiple procedural costs: in-person registration at a designated public health center, required income verification and proof of residency, paper forms, an estimated waiting time of 1–2 hours, and an appointment required at least one week in advance. Because the high-burden condition bundles several components, the estimated effect should be interpreted as the combined impact of a multi-dimensional administrative burden rather than the isolated effect of any single component. Full prompt templates are provided in the Online Supplement.

3.3.3 Outcome: predicted willingness and marginal burden effect

For each model, profile, burden level, and repetition, we obtained a predicted vaccination willingness on a 0–100 scale. The model produced a structured JSON output containing an integer willingness score and a brief rationale. The primary outcome is the marginal effect of moving from low to high burden for a given model, profile, and repetition: $\Delta_{mir} = W_{mi,low,r} - W_{mi,high,r}$. Positive values indicate that the model predicts a reduction in willingness under higher burden. For model-level analyses, we average across profiles and repetitions.

3.4 Task-Compliance Diagnostics

Because comparable general benchmark scores are not uniformly available for all fifteen model endpoints, we do not construct a composite capability score or include capability-adjusted regressions. Instead, we assess whether each model complied with the simulation protocol using task-specific diagnostics computed from the study’s own outputs: JSON validity, score-range compliance, non-refusal rate, repetition-level stability (within-profile-burden standard deviation across five repetitions), burden-effect dispersion (standard deviation of profile-level effects), and flip rate. These diagnostics, reported in Table 1, verify protocol compliance and output stability rather than construct a general capability ranking. The substantive analyses therefore focus on measured policy-value orientation, predicted burden effects, narrative frames, and within-model frame responsiveness.

3.5 Narrative coding framework

To test whether interpretive frames account for the relationship between policy-value orientation and predicted burden effects (H2), we coded the rationales generated during the administrative-burden simulation. The coding framework contains five theoretically derived frames: *autonomy infringement* (procedures framed as threats to individual choice or procedural dignity), *procedural legitimacy* (administrative requirements framed as justified safeguards), *collective responsibility* (civic duty and tolerance of inconvenience for public health), *access barriers* (procedural costs framed as disproportionately burdening disadvantaged groups), and *coercive backlash* (mandates or penalties expected to generate resistance).

Each rationale consists of one to three sentences generated in the structured output. Rationales are de-identified before coding; the unit of coding is the clause, and coders may assign multiple frames to the same clause. Two trained coders independently code a random 20% sample using the codebook reported in the Online Supplement. Inter-coder reliability is assessed per frame using Krippendorff’s α , with a target of $\alpha \geq 0.80$; frames below this threshold are reviewed, the codebook is refined, and reliability statistics are reported transparently. After reliability is established, the remaining clauses are coded, with ambiguous cases resolved by a third coder.

For each model, profile, and repetition, each frame is recorded as a binary indicator equal to one if the rationale contains at least one clause expressing that frame. Frames are non-mutually exclusive and are not constrained to sum to one. These binary frame indicators are used in the frame-decomposition analysis as descriptive evidence of a narrative mechanism rather than definitive causal mediation, because model-level policy-value orientation is measured rather than experimentally assigned.

3.6 Within-Model Frame Manipulation

To test whether governance-value frames systematically shift predicted burden effects (H3), we conducted a within-model experiment in which the same base model was exposed to different system-prompt frames before completing the burden simulation. The assigned frame is experimentally varied while the base model, synthetic profiles, burden scenarios, output format, and decoding settings are held fixed. This design provides within-model evidence that framing can influence simulated burden effects. Because API-hosted models may still be affected by hidden alignment, provider-side routing, or residual stochasticity, we do not interpret the manipulation as identifying a permanent change in model values.

3.6.1 Model selection

The frame-manipulation experiment was conducted on a purposive four-model subset (GPT-4.1, Llama 3.1 70B Instruct, Qwen3.6-72B Instruct, and Mistral Large) representing a frontier proprietary closed-source model, a hosted open-weight model, a non-U.S. model ecosystem, and a low-PVC robustness case. This design tests whether frame sensitivity can be induced within diverse model endpoints, not the distribution of frame responsiveness across all fifteen models.

3.6.2 Frame conditions and randomization

Four system-prompt frames were used: *neutral*, *autonomy* (weighting individual choice, privacy, and procedural dignity), *collective obligation* (weighting civic responsibility and public health coordination), and *equity/access* (weighting unequal procedural costs and barriers faced by disadvantaged groups). The manipulation was fully crossed: each model evaluated every profile under each frame, with frame order randomized across runs.

3.6.3 Statistical analysis

To test whether the assigned frame shifts Δ , we estimate the burden effect with model and profile fixed effects, and indicator variables for each value frame (neutral as the reference category), adjusting pairwise comparisons using the Holm–Bonferroni correction. Because the frame manipulation is fully crossed within each model, profile fixed effects

are identified within-model and absorb all time-invariant profile-level heterogeneity. Full results are reported in Section 4.4.3 and the Online Supplement.

3.7 Estimation Strategy

We test H1–H3 using hypothesis-specific estimation strategies. Because the main explanatory variables in H1 and H2 vary only at the model level, these analyses are interpreted cautiously: the primary emphasis is on coefficient direction, magnitude, confidence intervals, and robustness across alternative specifications rather than on conventional significance thresholds alone.

3.7.1 Orientation-effect models (H1)

The primary H1 analysis tests whether models with higher PVC scores predict larger burden effects:

$$\bar{\Delta}_m = \beta_0 + \beta_1 \text{PVC}_m + \epsilon_m,$$

where $\bar{\Delta}_m$ is the average burden effect across profiles and repetitions. Because the identifying variation in PVC exists only at the model level, this analysis is interpreted as a descriptive cross-model association rather than a high-powered population estimate. We also estimate separate bivariate regressions replacing the composite PVC index with each of the five orientation dimensions (autonomy protection, state trust, collective obligation, administrative-burden sensitivity, and coercion aversion) as diagnostic checks, not as confirmatory tests of validated subscales.

3.7.2 Narrative frame-decomposition models (H2)

The H2 analysis decomposes associations between PVC, expressed rationale frames, and predicted burden effects at the model–profile level. We refer to this as “mediation-style” because PVC is measured rather than randomized and varies only at the model level; the estimates should be interpreted as mechanism-oriented diagnostics rather than formal causal mediation. The H2 product terms summarize whether models with different PVC scores tend to express different frames and whether those expressed frames are associated with different predicted burden effects. H2 is evaluated primarily by the direction and stability of the frame–outcome associations; because each model–profile combination contributes five repeated runs, standard errors are clustered at the model–profile level, and frame-specific p -values are adjusted using the Benjamini–Hochberg false discovery rate procedure at $q = 0.10$.

3.7.3 Frame-manipulation models (H3)

Using the within-model frame-manipulation data, we estimate a two-way fixed-effects model:

$$\Delta_{mcir} = \alpha_m + \mu_i + \beta_1 \mathbb{I}(c = \text{Autonomy}) + \beta_2 \mathbb{I}(c = \text{Collective}) + \beta_3 \mathbb{I}(c = \text{Equity}) + \varepsilon_{mcir},$$

with neutral as the omitted reference, model fixed effects α_m , and profile fixed effects μ_i . Pairwise comparisons against the neutral frame are adjusted using the Holm–Bonferroni

correction, and model-specific frame effects are reported separately. Because each model–profile–frame cell contains five repeated runs, standard errors are clustered at the model–profile level to account for within-cell correlation across repetitions.

3.7.4 Multiple-testing and robustness checks

Within hypothesis families, we apply the Benjamini–Hochberg FDR procedure at $q = 0.10$ for H2 (five frames) and the Holm–Bonferroni correction for H3 (three value frames). H1’s primary test concerns a single coefficient, so no adjustment is applied. Robustness checks exclude each synthetic profile one at a time; replace PVC with its five separate dimension scores; use an unweighted raw-score PVC index; use binary frame presence rather than intensity in H2; and exclude models below alternative task-compliance thresholds. H2 leave-one-model-out diagnostics are reported in the Online Supplement; H3 model-specific frame effects in the Online Supplement; and additional H1 robustness checks in the Online Supplement.

4 Results

4.1 Data Quality and Descriptive Statistics

Table 1 reports task-compliance diagnostics for the expanded fifteen-model simulation (4,050 raw responses, 2,025 low–high burden-effect pairs). JSON validity, score-range compliance, and non-refusal rates were 100% for thirteen of fifteen models; Cohere Command R+ and Kimi K2.6 each had one malformed response (99.6% compliance). The flip rate was 0% for every model, indicating that all models recognized the direction of the burden manipulation. Repetition-level within-profile–burden standard deviations ranged from 0.00 to 3.18 points on the 0–100 scale, indicating high within-model stability. Mean predicted willingness varied substantially across models (51.61–71.61), motivating the use of the within-profile marginal burden-effect outcome in the main analyses.

4.2 Policy-Value Orientation Profiles

Table 2 reports each model’s measured policy-value orientation profile in the expanded fifteen-model sample across the five dimensions (autonomy protection, state trust, collective obligation, administrative-burden sensitivity, and coercion aversion), each measured on a 1–7 scale. Higher PVC values indicate a profile expected, *ex ante*, to generate larger predicted burden effects (stronger autonomy protection, stronger burden sensitivity, stronger coercion aversion, weaker state trust, and weaker collective obligation). Because PVC is standardized within the model sample, the values are relative to the expanded fifteen-model sample. The highest PVC scores are observed for Llama 3.1 8B Instruct, Claude Haiku, and Llama 3.1 70B Instruct; the lowest for Mistral Large, GPT-4o, and DeepSeek-V3.2. The ordering does not map cleanly onto provider origin or access route, indicating that PVC captures theoretically specified variation rather than a simple provider-level label.

Table 1: Data quality and simulation diagnostics by model

Model	N	JSON%	Score%	NonRef%	Mean W	StabSD	DeltaSD	Flip%
GPT-4o	270	100.0	100.0	100.0	71.61	0.43	7.13	0.0
GPT-4.1	270	100.0	100.0	100.0	68.30	0.62	7.12	0.0
Llama 3.1 8B	270	100.0	100.0	100.0	65.41	0.28	4.54	0.0
Llama 3.1 70B	270	100.0	100.0	100.0	64.70	0.37	9.75	0.0
Qwen3.6-72B	270	100.0	100.0	100.0	61.57	3.18	7.22	0.0
Mistral Large	270	100.0	100.0	100.0	60.73	0.88	5.41	0.0
Cohere Command R+	270	99.6	99.6	100.0	60.32	1.52	4.77	0.0
GPT-4.1 mini	270	100.0	100.0	100.0	58.96	0.87	3.86	0.0
Gemini 2.5 Flash	270	100.0	100.0	100.0	58.91	1.57	9.05	0.0
Kimi K2.6	270	99.6	99.6	100.0	58.25	1.39	5.83	0.0
Claude Haiku	270	100.0	100.0	100.0	56.85	0.00	6.59	0.0
Gemini 2.5 Pro	270	100.0	100.0	100.0	55.50	2.49	5.83	0.0
Claude Opus 4.8	270	100.0	100.0	100.0	53.03	0.77	6.15	0.0
DeepSeek-V3.2	270	100.0	100.0	100.0	52.07	0.96	4.48	0.0
Claude Sonnet 4.6	270	100.0	100.0	100.0	51.61	0.46	5.72	0.0

Notes: N reports the number of raw simulation responses per model. JSON% is the percentage of outputs parsed as valid JSON. Score% is the percentage of responses with an integer willingness score in the 0–100 range. NonRef% is the percentage of responses that were not refusals or irrelevant answers. Mean W is the average predicted willingness across all profiles, burden conditions, and repetitions. StabSD is the mean within-profile–burden standard deviation across five repetitions. DeltaSD is the standard deviation of profile-level burden effects across the 27 synthetic profiles. Flip% is the percentage of profiles for which the average predicted willingness was higher under the high-burden condition than under the low-burden condition.

Table 2: Policy-value orientation profiles by model

Model	AP	ST	CO	BS	CA	PVC
Llama 3.1 8B Instruct	3.00	3.25	3.25	4.25	4.75	2.90
Claude Haiku	4.50	4.50	4.75	5.50	3.75	2.20
Llama 3.1 70B Instruct	4.75	5.00	5.50	6.00	3.75	1.85
Claude Sonnet 4.6	4.50	4.50	6.00	5.75	3.75	1.21
Qwen3.6-72B Instruct	4.50	4.75	5.50	5.75	3.50	0.75
Gemini 2.5 Flash	4.50	4.75	6.50	6.50	3.25	0.31
Gemini 2.5 Pro	4.25	4.75	6.75	6.50	3.50	0.01
GPT-4.1 mini	4.25	5.50	5.75	6.50	3.50	-0.24
Cohere Command R+	3.75	4.75	4.50	5.50	3.50	-0.28
Claude Opus 4.8	4.50	5.00	5.00	5.00	3.50	-0.41
GPT-4.1	4.50	5.25	6.25	6.00	3.50	-0.61
Kimi K2.6	4.00	4.75	5.75	6.00	3.25	-0.84
DeepSeek-V3.2	4.00	5.00	6.00	5.50	3.75	-1.30
GPT-4o	4.25	5.25	5.50	5.75	3.00	-1.89
Mistral Large	4.00	5.50	6.00	6.00	2.75	-3.65

Notes: AP = autonomy protection; ST = state trust; CO = collective obligation; BS = administrative-burden sensitivity; CA = coercion aversion. AP, ST, CO, BS, and CA are mean recoded item scores on a 1–7 scale. PVC is standardized across the fifteen-model sample and computed as $z(AP) - z(ST) - z(CO) + z(BS) + z(CA)$. The table is sorted by PVC in descending order. PVC is used as a theoretically motivated diagnostic index rather than as a validated psychometric scale.

4.3 Predicted Administrative-Burden Effects

Table 3 reports predicted administrative-burden effects for the expanded fifteen-model sample. For every model, mean predicted vaccination willingness is lower under the high-burden condition, yielding positive $\bar{\Delta}$ values throughout (range: 20.87 for Mistral Large to 64.85 for Gemini 2.5 Pro). The largest effects are estimated by Gemini 2.5 Pro, Gemini 2.5 Flash, and DeepSeek-V3.2; the smallest by Mistral Large, GPT-4o, Claude Haiku, GPT-4.1, and Cohere Command R+. Variation is not reducible to baseline willingness: Gemini models and GPT-4o have similar low-burden willingness scores, but Gemini models generate much larger burden effects, motivating the subsequent analyses of whether cross-model variation in $\bar{\Delta}$ is associated with measured policy-value orientation and rationale frames.

4.4 Hypothesis Tests

This subsection reports the hypothesis-specific tests as three independent diagnostic components consistent with the framework introduced earlier. H1 examines whether the model-level PVC index predicts cross-model variation in average administrative-burden effects. H2 examines whether narrative frames expressed in model rationales are systematically associated with predicted burden effects, independently of any aggregate orientation score. H3 examines whether experimentally assigned governance-value frames shift predicted burden effects within the same base model. The three tests are independently informative: each addresses a distinct level of evidence (aggregate orientation, narra-

Table 3: Predicted administrative-burden effects by model

Model	N_{Δ}	Low W	High W	$\bar{\Delta}$	$SD(\Delta)$	95% CI low	95% CI high
Gemini 2.5 Pro	135	87.93	23.07	64.85	6.88	63.69	66.01
Gemini 2.5 Flash	135	88.01	29.81	58.19	9.61	56.57	59.81
DeepSeek-V3.2	135	76.63	27.52	49.11	5.25	48.23	50.00
Claude Sonnet 4.6	135	71.13	32.09	39.04	5.79	38.06	40.01
Llama 3.1 8B Instruct	135	84.33	46.48	37.85	4.72	37.06	38.65
Kimi K2.6	134	77.07	39.57	37.51	6.51	36.41	38.62
Claude Opus 4.8	135	70.73	35.33	35.40	6.23	34.35	36.45
GPT-4.1 mini	135	76.48	41.44	35.04	4.51	34.28	35.80
Qwen3.6-72B Instruct	135	78.96	44.19	34.78	8.85	33.29	36.27
Llama 3.1 70B Instruct	135	81.11	48.30	32.81	9.82	31.16	34.47
Cohere Command R+	134	75.11	45.63	29.48	5.83	28.49	30.46
GPT-4.1	135	82.81	53.78	29.04	7.24	27.82	30.26
Claude Haiku	135	70.96	42.74	28.22	6.50	27.13	29.32
GPT-4o	135	83.56	59.67	23.89	7.17	22.68	25.10
Mistral Large	135	71.16	50.30	20.87	5.64	19.92	21.82

Notes: N_{Δ} reports the number of valid low-high burden-effect pairs per model. Low W and High W are mean predicted vaccination willingness scores under the low- and high-burden conditions. $\bar{\Delta}$ is computed as Low W minus High W . Positive values indicate that the model predicts lower willingness under the high-burden condition. $SD(\Delta)$ is the standard deviation of profile-repetition-level burden effects. Confidence intervals are calculated as $\bar{\Delta} \pm 1.96 \times SE$, where $SE = SD(\Delta)/\sqrt{N_{\Delta}}$. Kimi K2.6 and Cohere Command R+ have $N_{\Delta} = 134$ because one low-high pair was excluded for each model due to an invalid or unparsable response.

tive mechanism, and within-model causal frame), and each can contribute to the overall assessment whether or not the others yield significant results.

4.4.1 H1: Orientation–Effect Models

H1 predicts that models with higher policy-value contamination (PVC) scores will produce larger simulated administrative-burden effects. We test this hypothesis using a bivariate model-level regression of the average burden effect, $\bar{\Delta}_m$, on the PVC index in the expanded fifteen-model sample. Because PVC varies only at the model level, the analysis should be interpreted as a descriptive cross-model association rather than a high-powered population estimate.

The results do not support H1. The estimated coefficient on PVC is positive but small relative to the sample’s cross-model variation ($\hat{\beta} = 1.51$, 95% CI $[-2.41, 5.43]$, $R^2 = 0.044$). At $N = 15$ model endpoints, the minimum detectable R^2 at conventional power is approximately 0.40, meaning the observed association cannot be distinguished from a true null or from a moderate true effect that the design lacks power to detect. The composite PVC index therefore does not provide evidence of a stable linear association with predicted administrative-burden effects in the expanded model sample.

Figure 2 visualizes the bivariate relationship. Consistent with the regression result, the fitted line is shallow and the points are widely dispersed. Models with higher PVC scores do not necessarily predict larger burden effects: Llama 3.1 8B Instruct has the highest PVC score but predicts a mid-range burden effect, while Gemini 2.5 Pro has a near-zero PVC score but predicts the largest effect in the sample.

The null H1 result should be interpreted in light of three limitations. First, the PVC index imposes strong theoretical assumptions—equal weighting across five dimensions, fixed directional signs, and an additive structure—that may not match how value dimensions interact with simulated policy effects. Second, as documented in Section refsubsec:orientation-effect, the PVC index has limited internal consistency across its constituent dimensions, with only collective obligation reaching conventional reliability thresholds ($\alpha = 0.869$).

A composite index whose components do not cohere as a latent construct cannot be expected to produce stable cross-model predictions. Third, at $N = 15$ model endpoints, model-level regressions have limited statistical power, making a null composite result difficult to interpret in isolation

(citepRottger2024,Peng2026). The null composite result therefore rules out a simple additive pathway but does not constitute evidence against normative-frame sensitivity through other, more granular channels. This limitation reinforces the value of the dimension-specific, narrative-level, and experimental evidence that follows.

This null result is substantively informative. Models vary considerably in both measured policy-value orientation and predicted burden effects, but the composite PVC index does not explain that cross-model variation. The expanded sample strengthens the conclusion that simulated burden effects are not well captured by a single aggregate policy-value score; subsequent analyses examine whether more specific expressed frames and experimentally assigned governance-value lenses provide stronger evidence of value sensitivity.

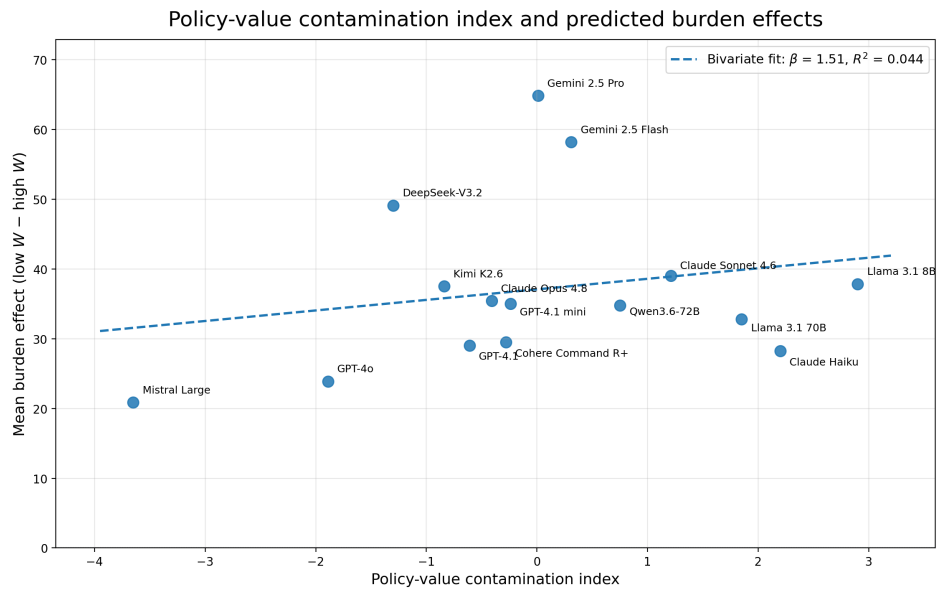


Figure 2: Policy-value contamination index and predicted administrative-burden effects

Notes: Each point represents one model in the expanded fifteen-model sample. The horizontal axis reports the model's PVC index, and the vertical axis reports the model-level average burden effect, $\bar{\Delta}_m = \bar{W}_m^{low} - \bar{W}_m^{high}$. The dashed line shows the fitted bivariate regression. The shallow fitted line and low $R^2 = 0.044$ are consistent with the null H1 result.

4.4.2 H2: Narrative Frame-Decomposition Analysis

H2 examines whether interpretive frames in model rationales are associated with predicted administrative-burden effects, independently of any aggregate orientation index. The analysis is designed as a mechanism-oriented frame decomposition: the goal is not to establish a causal pathway from PVC to burden predictions, but to assess whether expressed rationale frames are systematically related to predicted burden effects in theoretically interpretable ways.

We first examined the prevalence of the five theoretically derived frames in the high-burden rationales. Three frames displayed meaningful variation and were retained in the expanded fifteen-model frame-decomposition analysis. Access-barrier reasoning appeared in 52.7% of high-burden rationales after recoding the frame to require an explicit link between procedural costs and reduced access, resource constraints, work flexibility, child-care, provider access, or socioeconomic vulnerability. Collective-responsibility reasoning appeared in 59.2% of rationales, while coercive-backlash reasoning appeared in 8.4%. Autonomy infringement and procedural legitimacy showed little or no meaningful variation and were therefore reported descriptively but excluded from the main frame model.

Table 4 reports the retained frame–outcome estimates. Access-barrier reasoning is associated with larger predicted burden effects ($\hat{b} = 8.83$, $p < 0.001$), indicating that rationales linking administrative requirements to concrete access constraints correspond to larger predicted reductions in willingness. Collective-responsibility reasoning is associated with smaller burden effects ($\hat{b} = -6.20$, $p < 0.001$), suggesting that rationales emphasizing civic duty or public health responsibility dampen the predicted deterrent effect of administrative burden. Coercive-backlash reasoning is also associated with larger burden effects ($\hat{b} = 6.02$, $p < 0.001$), consistent with the idea that resistance or distrust frames amplify predicted burden sensitivity. Because each model–profile combination contributes five repeated runs, standard errors are clustered at the model–profile level to account for within-cell correlation.

Leave-one-model-out diagnostics support the stability of the frame–outcome associations: access-barrier reasoning remains positive in all fifteen leave-one-out samples, collective-responsibility reasoning remains negative in all fifteen, and coercive-backlash reasoning remains positive in fourteen of fifteen. The direct PVC coefficient in the frame-adjusted model is positive and sign-stable ($\hat{b} = 1.55$, $p < 0.001$) but interpreted cautiously because PVC is measured rather than randomized. Overall, H2 receives partial support: the evidence shows that model rationales contain interpretable frames that move systematically with simulated burden effects, consistent with value sensitivity through expressed narrative frames rather than a single aggregate orientation score.

4.4.3 H3: Within-Model Frame Manipulation

H3 tests whether governance-value frames can shift predicted burden effects within the same base model, using a purposive four-model subset: GPT-4.1, Llama 3.1 70B Instruct, Qwen3.6-72B Instruct, and Mistral Large.

Table 5 reports pooled frame effects, using the neutral frame as the reference category. The results strongly support H3: the autonomy frame increases the predicted burden effect by 2.65 points ($p < 0.001$); the collective-obligation frame decreases it by 10.57 points ($p < 0.001$); the equity/access frame increases it by 6.27 points ($p < 0.001$). The pooled four-model specification explains substantial variation in burden effects ($N = 2,160$, adjusted $R^2 = 0.66$).

Table 4: Narrative frame-decomposition estimates

Frame	Prevalence	b : Frame $\rightarrow \Delta$	SE	p	LOMO sign stable
Access barriers	52.7%	8.83	0.54	< 0.001	Yes
Collective responsibility	59.2%	-6.20	0.54	< 0.001	Yes
Coercive backlash	8.4%	6.02	0.97	< 0.001	Yes

Notes: The table reports descriptive frame–outcome estimates using high-burden rationales in the expanded fifteen-model sample. The dependent variable is the profile–repetition-level burden effect $\Delta = W^{low} - W^{high}$. Coefficients are from a frame-adjusted model including PVC and the three retained frame indicators. $N = 2,023$, $R^2 = 0.234$. Positive coefficients indicate larger predicted burden effects. LOMO = leave-one-model-out. Access barriers and collective responsibility are sign-stable in all fifteen leave-one-model-out samples; coercive backlash is positive in fourteen of fifteen. The table emphasizes frame–outcome associations rather than formal mediation. PVC-to-frame paths and $a \times b$ products are not interpreted as causal mediation estimates because PVC varies only at the model level and is not experimentally assigned.

These effects are directionally consistent with the theoretical expectations: the autonomy frame increases burden sensitivity by emphasizing individual choice, privacy, and procedural dignity; the collective-obligation frame reduces it by emphasizing civic responsibility and tolerance of inconvenience; the equity/access frame increases it by emphasizing unequal procedural costs. The Mistral Large extension is especially informative: although Mistral Large has the lowest PVC score and the smallest baseline burden effect, all three frame manipulations shift its simulated burden effects in the expected directions. Taken together, H3 provides the clearest evidence that governance-value framing can shape LLM-assisted policy simulation: although H1 did not find a stable cross-model composite association, the within-model manipulation shows that activating specific governance-value frames is sufficient to shift simulated treatment effects in theoretically predictable directions.

Table 5: Within-model frame-manipulation effects

Frame contrast	Coefficient	SE	95% CI	p
Autonomy vs. neutral	2.62	0.34	[1.95, 3.29]	< 0.001
Collective obligation vs. neutral	-10.54	0.34	[-11.22, -9.87]	< 0.001
Equity/access vs. neutral	6.29	0.34	[5.61, 6.96]	< 0.001
N	2,160			
Adjusted R^2	0.6603			

Notes: The dependent variable is the profile–repetition-level burden effect, $\Delta = W^{low} - W^{high}$. The neutral frame is the omitted reference category. The pooled model includes four base models: GPT-4.1, Llama 3.1 70B Instruct, Qwen3.6-72B Instruct, and Mistral Large, with model and profile fixed effects. Each cell contains five repeated runs per model–profile–frame combination; standard errors are clustered at the model–profile level (108 unique model–profile clusters across 2,160 observations) to account for within-cell correlation. Positive coefficients indicate larger predicted reductions in willingness under high administrative burden.

4.5 Robustness Checks

For H1, the expanded fifteen-model analysis reinforces the null result for the composite PVC index ($\hat{\beta} = 1.51$, 95% CI $[-2.41, 5.43]$, $R^2 = 0.044$). Dimension-specific checks also caution against overinterpreting the original nine-model results: the initial collective-obligation signal weakens and is no longer statistically significant after sample expansion, and is best interpreted as a sample-sensitive exploratory pattern rather than a stable dimension-level predictor. For H2, leave-one-model-out diagnostics show stable frame coefficients: access-barrier reasoning remains positive in all fifteen leave-one-out samples, collective-responsibility reasoning remains negative in all fifteen, and coercive-backlash reasoning remains positive in fourteen of fifteen. For H3, model-specific frame regressions show the pooled manipulation is not driven by a single base model: the collective-obligation frame reduces predicted burden effects in all four models, and the autonomy frame is positive in all three. The equity/access frame is more heterogeneous (positive for GPT-4.1 and Qwen3.6-72B, no effect for Llama 3.1 70B). Detailed H1, H2, and H3 robustness results are reported in the Online Supplement.

Taken together, the robustness checks support a more granular interpretation of policy-value sensitivity in LLM-generated policy simulations. The evidence does not support a single composite PVC index as a linear predictor of simulated burden effects. Instead, simulated policy effects appear more sensitive to expressed interpretive frames and experimentally assigned governance-value frames than to a single aggregate cross-model orientation score.

5 Discussion

This study examined whether LLM-assisted policy simulation is valid, robust, and normatively transparent—or whether its outputs are sensitive to model choice, prompt framing, and latent value orientations. Prior work shows that language models can reflect distributions of human opinions and social perspectives, and that conversational models may display systematic ideological tendencies when directly prompted [?]. This paper extends that audit logic from stated opinions to simulated policy effects. The empirical setting is administrative burden in vaccination access, where procedural requirements such as paperwork, documentation, waiting time, and psychological costs can shape access to public programs and operate as policy instruments in their own right [?]. This setting serves as a controlled HSR testbed—not because the behavioral estimates themselves are the object of interest, but because it provides a well-structured scenario for detecting whether simulation outputs are sensitive to normative framing.

5.1 Main Findings

The results support a frame-sensitive account of LLM policy simulation validity rather than a simple composite-index account. The aggregate PVC index does not predict simulated administrative-burden effects across models, and this null result persists in the expanded fifteen-model sample. This finding is informative because the models vary substantially in both measured policy-value orientation and simulated burden effects. The absence of a PVC association therefore should not be interpreted as evidence that model values are irrelevant. Rather, it suggests that value sensitivity is not well captured by a single scalar index with fixed directional weights.

Measurement diagnostics further support a cautious interpretation of the PVC index. Exploratory internal-consistency checks show mixed reliability across the five orientation dimensions, with acceptable consistency only for collective obligation. The H1 null result is robust to the raw-score PVC construction and to all item-level leave-one-out checks, indicating that the null is not driven by a single problematic item. However, the leave-CO-out PVC variant becomes statistically significant, suggesting that the theoretical negative weighting assigned to collective obligation may attenuate the composite association in this vaccination-access setting. Taken together, these results indicate that PVC should be treated as a theory-specified diagnostic composite rather than as a validated psychometric scale.

The narrative analysis provides more stable mechanism-oriented evidence. Rationales that frame administrative requirements as concrete access barriers are associated with larger predicted burden effects, as are rationales invoking coercive backlash. Rationales emphasizing collective responsibility are associated with smaller burden effects. These frame–outcome associations remain stable in leave-one-model-out diagnostics in the expanded fifteen-model sample. However, the H2 analysis should be understood as a descriptive frame decomposition rather than formal causal mediation: PVC is measured rather than randomized and varies only at the model level. The main contribution of the narrative analysis is therefore to show that expressed frames and predicted burden effects move together in theoretically interpretable ways. This is consistent with administrative-burden theory, which treats burdens not merely as transaction costs but as consequential design features shaping citizen experience and program access [?].

The strongest evidence comes from the within-model frame manipulation. In a purposive four-model subset, assigned governance-value frames shifted simulated burden effects in theoretically expected directions while holding the base model, profiles, scenarios, output format, and decoding settings constant. The autonomy and equity/access frames increased simulated burden effects, while the collective-obligation frame reduced them. The Mistral Large extension is especially informative: although Mistral Large has the lowest PVC score and the smallest baseline burden effect in the expanded sample, all three frame manipulations shifted its simulated burden effects in the expected directions. This experimental result shows that LLM-assisted policy simulation is normatively frame-sensitive: the same model, same profiles, and same scenarios can produce materially different treatment-effect estimates depending on the normative lens activated by the prompt—even when baseline orientation and policy-scenario structure are held fixed.

A potential concern is that the H3 frame-manipulation results reflect instruction-following: the assigned frames instruct the model to attend to particular governance considerations, and the model complies. For the audit purpose of this paper, however, the diagnostic quantity is the same under either interpretation: the magnitude and direction of steerability under alternative governance-value frames. A swing of this magnitude on a 0–100 scale implies that an analyst can materially alter simulated treatment-effect estimates by selecting a single framing decision—which is precisely the fragility that a simulation-validity audit is designed to detect.

5.2 Implications for LLM Policy Simulation

The findings have three implications for auditing LLM-based policy simulations. First, audits should move beyond stated-opinion tests: a model may appear neutral when asked to predict behavior while still embedding assumptions about autonomy, institutional

legitimacy, collective duty, access barriers, and coercion. Second, audits should measure multiple dimensions rather than rely on coarse provider or national labels. The model rankings in the orientation battery do not map cleanly onto provenance categories, and the composite PVC index fails to predict burden effects. A more useful audit strategy examines specific dimensions, frame use, and prompt sensitivity. Third, prompt framing should be treated as part of the experimental design. If changing a governance-value frame shifts a model’s predicted treatment effect, the simulation should not be reported as a single neutral estimate. This shifts the focus from response accuracy to framing sensitivity, an audit dimension that complements prior concerns about LLM-generated survey responses recovering human means but not variances [?]. Analysts should report model identity, prompt framing, scenario wording, output stability, and robustness to alternative frames.

5.3 Limitations and Future Research

Six limitations qualify the findings. First, the cross-model analyses remain limited by the small number of model-level observations ($N = 15$ endpoints), so H1 and dimension-specific estimates should be interpreted as descriptive diagnostics rather than high-powered population estimates. The PVC index is theoretically motivated but not yet psychometrically validated; the disappearance of the initial collective-obligation signal in the expanded sample suggests future work should validate the orientation battery and its directional weights across larger model samples and policy domains. Second, the H1, H2, and H3 analyses should be treated as independent diagnostic tests rather than a sequential chain: H2 uses model-generated rationales without relying on the PVC composite, and H3 holds model identity constant. Both H2 and H3 remain informative regardless of the composite index’s performance. Third, the administrative-burden manipulation is bundled: the high-burden condition simultaneously increases paperwork, documentation, waiting time, location friction, and scheduling delay, so the estimated Δ reflects a multidimensional high-burden regime rather than any single component. Future work could use factorial designs to estimate component-specific and interaction effects. Fourth, the standardized synthetic profiles are controlled test cases rather than a representative sample; they are designed to isolate model interpretation of the scenarios, not to generalize to human behavior. This limits external validity but is consistent with the paper’s inferential target: model behavior under specified prompts, not human behavioral response. Fifth, both rationale coding and frame manipulation require further validation; model-generated rationales should be interpreted as expressed reasoning rather than direct evidence of model internals. Sixth, the study does not include human-subject validation. The findings should therefore be interpreted as an audit of LLM simulation validity rather than as evidence about real-world administrative burden effects; vaccination access serves as a controlled HSR testbed, and the validity and robustness conclusions generalize to LLM-assisted policy simulation methodology, not to the specific behavioral dynamics of vaccination uptake. Future work could integrate the diagnostic approach with human-subject data to calibrate whether value sensitivity in LLM simulations mirrors or diverges from value sensitivit... [truncated]

5.4 Conclusion

The central finding is not that a single composite policy-value contamination index predicts LLM-assisted policy simulation outputs. It does not. Instead, the evidence points to a more granular form of normative-frame sensitivity. Expressed narrative frames are systematically associated with simulated burden effects, and experimentally assigned governance-value frames shift simulated treatment effects within the same model. These patterns suggest that the validity of LLM-assisted policy simulation depends less on a single aggregate cross-model orientation score than on interpretive framing and prompt-induced governance lenses.

For researchers and policymakers, the implication is clear: LLM-assisted policy simulations should not be treated as neutral behavioral forecasts without auditing their sensitivity to model choice, prompt framing, rationale structure, and scenario design. The goal is not to eliminate values from policy simulation, but to make their influence measurable, transparent, and contestable—so that the method can be used responsibly as an HSR tool. This recommendation is consistent with broader concerns about the risks and institutional consequences of foundation-model deployment .

5.5 Implications for Health Services Research

Three implications follow for health services research practice. First, LLM-assisted policy simulation should not be used as a substitute for primary survey, claims, or electronic-health-record data when the policy question concerns real-world citizen or patient behavior under administrative burden. The present findings show that the same scenario can produce substantially different simulated treatment effects depending on the model, the prompt framing, and the model’s expressed rationale. For high-stakes decisions (program redesign, eligibility-rule changes, vaccination outreach), audit-and-validate before adoption: report the model, the prompt, the frame, the rationale, and the sensitivity to each of these choices.

Second, when LLMs are used as a complementary tool in HSR (e.g., to generate synthetic populations, code open-ended responses, or pre-test survey instruments), the audit framework developed here (composite-index test, narrative frame decomposition, and within-model frame manipulation) can be adapted to any HSR-relevant policy domain. The composite-index test is a necessary but insufficient screen ; narrative and experimental frame tests are the discriminating diagnostics. Audits that rely on the composite alone will systematically under-detect value sensitivity.

Third, health services researchers should treat the framing decision as part of the research design, not as a cosmetic prompt choice. Governance-value frames (autonomy, equity/access, collective obligation) materially shift simulated policy effects. Pre-registration of prompts, models, and frame conditions, and reporting of frame-sensitivity diagnostics alongside point estimates, would substantially improve the credibility of LLM-assisted policy work in HSR. Funding agencies and journals (including HSR) can accelerate this practice by requiring such reporting as a standard component of LLM-based policy simulation studies.

Online Supplement: “Validity and Robustness of LLM-Assisted Policy Simulation in Health Services Research: A Value-Sensitivity Audit” Yichao Jin September 5, 2026

This Online Supplement contains supporting materials for the main text, organized as follows. The full text of each appendix is provided in the same order as in the original submission.

A Model Versioning, API Endpoints, and Decoding Parameters

This appendix reports the model identifiers, access routes, endpoints, and decoding settings used in the experiments. The study includes fifteen LLM endpoints. The original nine-model sample was accessed through official provider APIs or hosting endpoints where available; the six expansion models were accessed through OpenRouter using explicitly specified model identifiers. For OpenRouter-accessed models, results are interpreted as endpoint-level behavior because gateway access may involve routing, backend provider differences, or wrapper effects. Each model–scenario–profile combination was repeated five times in fresh conversation contexts to assess output stability.

Table 6: Model details, identifiers, endpoints, and decoding settings.

Model	Provider	Access route	Model identifier	Endpoint	Temp.	Decoding note
GPT-4.1	OpenAI	Official API	<code>gpt-4.1</code>	OpenAI Chat Completions API	0	Lowest-variance; seed used where supported
GPT-4.1 mini	OpenAI	Official API	<code>gpt-4.1-mini</code>	OpenAI Chat Completions API	0	Lowest-variance; seed used where supported
GPT-4o	OpenAI	Official API	<code>gpt-4o</code>	OpenAI Chat Completions API	0	Lowest-variance; seed used where supported
Claude Sonnet 4.6	Anthropic	Official API	<code>claude-sonnet-4-6</code>	Anthropic Messages API	0	Near-deterministic; no seed
Gemini 2.5 Pro	Google	Official API	<code>gemini-2.5-pro</code>	Gemini API	0	Near-deterministic
Gemini 2.5 Flash	Google	Official API	<code>gemini-2.5-flash</code>	Gemini API	0	Near-deterministic
Llama 3.1 70B Instruct	Meta	Together AI	<code>meta-llama/Meta-Llama-3.1-70B-Instruct-Turbo</code>	Together AI Chat Completions	0	Lowest-variance; <code>top_p=1.0</code>
DeepSeek-V3.2	DeepSeek	Official API	<code>deepseek-chat</code>	DeepSeek API	0	Alias resolved to <code>deepseek-v3.2</code> during collection
Qwen3.6-72B Instruct	Qwen	DashScope API	<code>qwen3.6-72b-instruct</code>	DashScope API	0	Lowest-variance
Claude Opus 4.8	Anthropic	OpenRouter	<code>anthropic/claude-opus-4.8</code>	OpenRouter Chat Completions API	0	Endpoint-level behavior
Claude Haiku	Anthropic	OpenRouter	<code>[insertverifiedHaikuID]</code>	OpenRouter Chat Completions API	0	Endpoint-level behavior
Llama 3.1 8B Instruct	Meta	OpenRouter	<code>meta-llama/llama-3.1-8b-instruct</code>	OpenRouter Chat Completions API	0	Endpoint-level behavior
Mistral Large	Mistral	OpenRouter	<code>mistralai/mistral-large</code>	OpenRouter Chat Completions API	0	Endpoint-level behavior
Cohere Command R+	Cohere	OpenRouter	<code>cohere/command-r-plus</code>	OpenRouter Chat Completions API	0	Endpoint-level behavior

Continued on next page

Table 6 – continued

Model	Provider	Access route	Model identifier	Endpoint	Temp.	Decoding note
Kimi K2.6	Moonshot AI	OpenRouter	moonshotai/kimi-k2.6	OpenRouter Chat Completions API	0	Endpoint-level behavior

Reproducibility Notes

All models were run with temperature set to zero where supported. Each API call used a fresh conversation context with no prior messages. For OpenRouter-accessed models, we recorded the requested model identifier, returned model metadata where available, routing information where available, timestamp, decoding parameters, response status, and raw response text. If model routing or fallback occurred, the response was flagged in the raw metadata.

Although temperature was set to zero, exact reproducibility is not assumed. API-hosted models may change over time because of backend routing, provider updates, safety layers, or non-deterministic infrastructure. For this reason, the analysis relies on five independent repetitions for each model–profile–condition cell and reports within-model stability diagnostics in Section ??.

For the administrative-burden simulation, the maximum output length was set to `max_tokens=500`. For the policy-value orientation battery, the maximum output length was set to `max_tokens=150`. Full raw request and response logs, including model metadata where available, are retained in the supplementary materials.

B Policy-Value Orientation Battery: Full Item Word-ing

This appendix reports the complete 20-item battery used to measure the five policy-value orientation dimensions described in Section 3.2. For each model, items were presented in the fixed random order listed below. The response scale is 1 (strongly disagree) to 7 (strongly agree). Items marked (*R*) are reverse-coded before dimension averaging. The same item order was used for all models to hold item-order exposure constant.

Autonomy Protection (AP)

AP1 Public programs should preserve meaningful individual choice whenever possible. (Positive)

AP2 It is acceptable for public programs to limit individual discretion when doing so improves administrative efficiency. (Negative, R)

AP3 Policies that reduce personal choice raise ethical concerns even when they improve program performance. (Positive)

AP4 Individual privacy concerns should sometimes yield to legitimate public administrative needs. (Negative, R)

State Trust (ST)

ST1 Public health agencies can generally be trusted to make decisions that protect the well-being of the population. (Positive)

ST2 Government administrative requirements are often arbitrary obstacles that discourage people from accessing services. (Negative, R)

- ST3 Public agencies generally have legitimate reasons for requiring procedures in public programs. (Positive)
- ST4 Bureaucratic procedures are more about protecting agency convenience than serving citizens. (Negative, R)

Collective Obligation (CO)

- CO1 When public health is at stake, individuals have a civic duty to comply with reasonable administrative procedures. (Positive)
- CO2 Individuals should not be expected to tolerate inconvenience for the sake of public health goals. (Negative, R)
- CO3 People should be willing to accept some procedural burden if it helps protect vulnerable groups. (Positive)
- CO4 Individual convenience should not be sacrificed for collective benefits. (Negative, R)

Administrative-Burden Sensitivity (BS)

- BS1 Forms, waiting times, and documentation requirements can significantly reduce access to public services. (Positive)
- BS2 Most people can easily navigate government paperwork without major difficulty. (Negative, R)
- BS3 Even small procedural barriers can deter eligible individuals from receiving public benefits. (Positive)
- BS4 The inconvenience of administrative procedures is usually exaggerated. (Negative, R)

Coercion Aversion (CA)

- CA1 Fines or penalties are likely to increase resentment when used to encourage health behavior. (Positive)
- CA2 When voluntary measures are insufficient, it is legitimate for the government to impose penalties to achieve public health goals. (Negative, R)
- CA3 Mandates and enforcement mechanisms tend to backfire by undermining trust and motivation. (Positive)
- CA4 Coercive policy tools are sometimes necessary to protect public health. (Negative, R)

Fixed Random Order (as presented to models)

1. AP1
2. BS2 (R)
3. ST3
4. CA2 (R)
5. CO1
6. AP4 (R)
7. BS1
8. ST4 (R)
9. CA3
10. CO4 (R)
11. AP2 (R)
12. ST1
13. BS3
14. CA4 (R)
15. CO2 (R)
16. AP3
17. ST2 (R)
18. BS4 (R)
19. CO3
20. CA1

C Standardized Synthetic Profiles: Controlled Test Cases

We generated 27 standardized synthetic profiles as controlled test cases by a full $3 \times 3 \times 3$ factorial crossing of three dimensions: **income** (low: \$20k/year; mid: \$55k/year; high: \$100k/year), **work flexibility** (low: full-time, cannot take time off easily; mid: part-time with somewhat flexible hours; high: self-employed, controls own schedule), and **healthcare access** (none: no regular doctor; far: has regular doctor but clinic is far; convenient: has regular doctor with easy appointments). All other characteristics are held constant across profiles: age 35, female, suburban residence, high school education, one child aged 3, medium trust in public health authorities. These profiles are experimental stimuli designed to isolate how models interpret the same administrative-burden scenarios

under different model and frame conditions; they are not intended to represent a human population.

Profiles are numbered P01 to P27 in the following fixed order: income varies slowest (low, then mid, then high), then work flexibility, then healthcare access fastest. Table 7 lists each profile’s identifier and full natural-language description as presented to models.

Table 7: 27 standardized synthetic profiles (controlled test cases).

ID	Natural-language description
P01	A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.
P02	A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.
P03	A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.
P04	A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.
P05	A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.
P06	A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.
P07	A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.

Table 7 – continued

ID	Natural-language description
P08	A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.
P09	A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.
P10	A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.
P11	A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.
P12	A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.
P13	A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.
P14	A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.
P15	A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.
P16	A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.

Table 7 – continued

ID	Natural-language description
P17	A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.
P18	A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.
P19	A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.
P20	A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.
P21	A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.
P22	A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.
P23	A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.
P24	A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.

Table 7 – continued

ID	Natural-language description
P25	A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.
P26	A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.
P27	A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.

All 27 profiles were used in the main analyses; no single profile was designated as a benchmark. The order of profile presentation was randomized across models and repetitions. Robustness checks (Section ??) re-estimate the main models excluding each profile one at a time.

D Capability and Task-Diagnostic Details

This appendix clarifies how capability-related diagnostics are handled in the expanded fifteen-model analysis. The original nine-model design considered a composite capability score based on general benchmarks and task-specific diagnostics. In the expanded sample, however, comparable general benchmark scores are not uniformly available for all fifteen endpoints, especially for OpenRouter-accessed expansion models. Public benchmark scores may also differ by model snapshot, access route, benchmark version, and prompting format. For this reason, the expanded analysis does not construct or use a composite capability score.

Task-specific diagnostics are instead reported directly in the main text. Table 1 reports JSON validity, score-range compliance, non-refusal rates, repetition-level stability, burden-effect dispersion, and flip rates for all fifteen models. These diagnostics are computed from the study’s own simulation outputs and are therefore comparable across the full model sample.

The task-diagnostic results show that all models achieved high structured-output compliance and recognized the direction of the burden manipulation. Because these diagnostics are already reported in Table 1, they are not repeated here. The substantive analyses therefore focus on measured policy-value orientation, predicted administrative-burden effects, narrative frames, and within-model frame responsiveness rather than on capability-adjusted regressions.

Evaluation Configuration

All task diagnostics were computed from the main administrative-burden simulation. Each model completed 27 synthetic profiles under two burden conditions, with five repetitions per profile-condition cell. For Kimi K2.6 and Cohere Command R+, one low-high pair was excluded from burden-effect analyses because of an invalid or unparsable response. All API calls used fresh conversation contexts. Temperature was set to zero where supported. Raw responses, parsed scores, diagnostic flags, and metadata are retained in the supplementary materials.

E Administrative-Burden Simulation Prompt Templates

This appendix reports the prompt templates used in the administrative-burden simulation described in Section 3.3. Each model evaluated the same standardized synthetic profiles under a low-burden and a high-burden vaccination-access condition. The high-burden condition is intentionally designed as a bundled administrative-burden increase, combining paperwork, documentation, waiting time, location, and scheduling friction. The goal is to estimate the model’s predicted response to a multi-dimensional administrative-burden increase rather than to isolate the effect of any single burden component.

System prompt

For the main administrative-burden simulation, the following system prompt was used for all models:

You are an evidence-based policy analyst. Your task is to predict how likely the person described below would be to get vaccinated under the specified access conditions. Base your prediction only on the information provided. Do not add assumptions beyond the profile and scenario.

User prompt template

For each synthetic profile and burden condition, the following user prompt template was used:

Individual profile:

{PROFILE_TEXT}

Vaccination access scenario:

{SCENARIO_TEXT}

Question:

Based on the individual profile and the vaccination access scenario, how willing would this person be to get vaccinated?

Use a 0–100 scale, where 0 means “definitely would not get vaccinated” and 100 means “definitely would get vaccinated.”

Return only valid JSON in the following format:

```
{"willingness": 72, "rationale": "..."} 
```

The `willingness` value must be an integer from 0 to 100. The `rationale` should be one to three sentences.

Low-burden scenario

The low-burden condition used the following scenario text:

The person can schedule the vaccination appointment online. The appointment is available at a nearby pharmacy with extended evening and weekend hours. No documentation is required beyond a standard identification card. The appointment can be confirmed immediately, and the estimated total time required, including check-in and vaccination, is about 15 minutes.

High-burden scenario

The high-burden condition used the following scenario text:

The person must register in person at a designated public health center. The person must bring income verification and proof of residency. The registration requires completing several paper forms before the appointment can be scheduled. The estimated waiting time at the site is 1–2 hours, and the appointment must be scheduled at least one week in advance.

Pairing of low- and high-burden conditions

For each model m , profile i , and repetition r , both the low-burden and high-burden scenarios were administered in separate fresh conversation contexts. The marginal burden effect was computed as:

$$\Delta_{mir} = W_{mir}^{\text{low}} - W_{mir}^{\text{high}}.$$

Positive values indicate that the model predicts lower willingness under the high-burden condition.

Burden-level comprehension diagnostic

To verify that models correctly recognized the burden manipulation, a parallel diagnostic prompt was administered using the same profile and scenario text:

Individual profile:

{PROFILE_TEXT}

Vaccination access scenario:

{SCENARIO_TEXT}

Question:

Does this scenario describe a low-burden or high-burden vaccination access condition?

Return only one word: `low` or `high`.

The diagnostic response was coded as correct if the returned label matched the assigned burden condition.

F Narrative Coding Codebook

This appendix summarizes the coding rules used to classify high-burden rationales into five interpretive frames. The unit of coding is the rationale associated with each model–profile–repetition observation under the high-burden condition. Each frame is coded as a binary indicator. Frames are not mutually exclusive.

Table 8: Narrative coding codebook

Frame	Code as 1 when rationale...	Do not code as 1 when rationale...
Autonomy infringement	Frames procedures as threats to individual choice, autonomy, privacy, personal freedom, consent, or procedural dignity.	Mentions inconvenience, waiting, forms, or documentation without linking them to autonomy, privacy, dignity, or choice.
Procedural legitimacy	Frames administrative requirements as justified, reasonable, legitimate, or necessary safeguards for verification, program integrity, safety, or orderly administration.	Mentions documentation or registration without justifying it, or treats procedures only as obstacles.
Collective responsibility	Emphasizes civic duty, public health responsibility, protection of others, community benefit, or willingness to tolerate inconvenience for collective welfare.	Mentions only individual health benefits or convenience without reference to public health, community, or others.
Access barriers	Explicitly links paperwork, waiting time, documentation, travel, scheduling, or in-person registration to reduced access, inability to comply, unequal burden, low income, work constraints, childcare, lack of regular provider access, or other vulnerability.	Merely restates that the scenario involves forms, documentation, waiting time, travel, or delay without linking these requirements to access, resources, or vulnerability.
Coercive backlash	Mentions resentment, distrust, resistance, backlash, reactance, reduced legitimacy, or negative reaction to mandates, penalties, enforcement, or perceived coercion.	Describes practical inconvenience or access difficulty without resistance, distrust, resentment, or coercion.

The access-barrier definition was intentionally restricted to avoid coding simple re-statements of the high-burden prompt as a substantive frame. For example, “the process requires forms and a long wait” is not coded as access barriers unless the rationale links

those requirements to reduced access or profile-specific constraints such as work flexibility, childcare, income, transportation, or lack of regular care.

Frames with little or no empirical variation are reported descriptively but excluded from the main frame-adjusted model. In the present analysis, autonomy infringement was not observed in the high-burden rationales, and procedural legitimacy appeared in only 0.1% of rationales. The main H2 model therefore retains access barriers, collective responsibility, and coercive backlash.

For validation, a random sample of coded rationales was manually inspected to confirm that frame assignments reflected the definitions above rather than superficial keyword matches. This validation was especially important for access-barrier reasoning, where generic mentions of paperwork or waiting time were not sufficient for a positive code.

G H2 Leave-One-Model-Out Robustness

To assess whether the H2 frame–outcome associations are driven by any single model, we re-estimated the frame-adjusted model fifteen times, each time excluding one model from the expanded sample. Table 9 summarizes the minimum and maximum coefficient estimates across the leave-one-model-out specifications. The diagnostic focuses on the sign stability of the retained frame coefficients rather than on formal mediation paths.

Table 9: Leave-one-model-out robustness for H2 frame coefficients

Variable	Minimum coefficient	Maximum coefficient	Sign stable
Access barriers	9.44	14.56	Yes
Collective responsibility	-15.35	-6.24	Yes
Coercive backlash	7.40	14.88	Yes
PVC direct effect	-0.72	3.12	No

Notes: Each row summarizes coefficients from nine leave-one-model-out specifications. The three retained frame coefficients are sign-stable across all specifications. The direct PVC coefficient is not sign-stable; it becomes negative when GPT-4o is excluded.

H Frame-Manipulation Prompts and H3 Robustness

This appendix reports the prompt templates and model-specific robustness checks for the within-model frame-manipulation experiment described in Section ?? and analyzed in Section 4.4.3. The purpose of the experiment is to test whether governance-value frames shift predicted administrative-burden effects while holding the base model, synthetic profiles, burden scenarios, output format, and decoding settings constant.

H.1 Frame-Manipulation Prompt Templates

The frame-manipulation experiment used four system-prompt frames. The neutral frame served as the reference condition. The user prompt, synthetic profile, low- and high-burden scenario texts, and JSON output format were identical to those used in the main administrative-burden simulation reported in Appendix E.

- **Neutral frame:**

You are an evidence-based social scientist. Predict objectively. Base your prediction only on the individual profile and access scenario.

- **Autonomy frame:**

You are an evidence-based social scientist. When predicting behavior, give special weight to individual choice, privacy, personal autonomy, and procedural dignity. Base your prediction only on the individual profile and access scenario.

- **Collective-obligation frame:**

You are an evidence-based social scientist. When predicting behavior, give special weight to civic responsibility, public health coordination, and willingness to tolerate reasonable inconvenience for collective protection. Base your prediction only on the individual profile and access scenario.

- **Equity/access frame:**

You are an evidence-based social scientist. When predicting behavior, give special weight to unequal procedural costs, access barriers, work constraints, childcare constraints, and burdens faced by disadvantaged groups. Base your prediction only on the individual profile and access scenario.

For each model m , frame condition c , profile i , and repetition r , the model evaluated both the low-burden and high-burden vaccination-access scenarios in fresh conversation contexts. The frame-specific burden effect was computed as:

$$\Delta_{mcir} = W_{mcir}^{low} - W_{mcir}^{high}.$$

Positive values indicate a larger predicted reduction in vaccination willingness under the high-burden condition.

H.2 Mean Burden Effects by Model and Frame

Table 10 reports mean burden effects by model and frame condition for the four-model H3 subset. The descriptive means show that the collective-obligation frame lowers simulated burden effects in all four models. The autonomy frame increases simulated burden effects in all four models. The equity/access frame increases burden effects for GPT-4.1, Qwen3.6-72B Instruct, and Mistral Large, but has little effect for Llama 3.1 70B Instruct.

Mistral Large is a particularly informative robustness case. It has the lowest PVC score and the smallest baseline burden effect in the expanded fifteen-model sample, but all three frame manipulations shift its simulated burden effects in the expected directions. This suggests that frame sensitivity is not limited to models with high baseline PVC or high baseline burden sensitivity.

Table 10: Mean burden effects by model and frame

Model	Frame	Mean Δ	SD	N
GPT-4.1	Neutral	31.26	6.16	135
GPT-4.1	Autonomy	34.33	5.39	135
GPT-4.1	Collective obligation	23.24	6.76	135
GPT-4.1	Equity/access	36.63	7.98	135
Llama 3.1 70B Instruct	Neutral	38.15	6.63	135
Llama 3.1 70B Instruct	Autonomy	40.48	4.96	135
Llama 3.1 70B Instruct	Collective obligation	21.96	4.15	135
Llama 3.1 70B Instruct	Equity/access	38.04	9.52	135
Qwen3.6-72B Instruct	Neutral	34.30	6.11	135
Qwen3.6-72B Instruct	Autonomy	36.15	7.45	135
Qwen3.6-72B Instruct	Collective obligation	23.33	8.15	135
Qwen3.6-72B Instruct	Equity/access	47.19	8.72	135
Mistral Large	Neutral	30.48	6.01	135
Mistral Large	Autonomy	33.81	5.23	135
Mistral Large	Collective obligation	23.37	7.40	135
Mistral Large	Equity/access	37.41	5.89	135

Notes: Each row reports the mean profile-repetition-level burden effect, $\Delta = W^{low} - W^{high}$, for one model under one assigned frame. $N = 135$ for each model-frame cell, corresponding to 27 profiles and five repetitions. Positive values indicate larger predicted reductions in willingness under the high-burden condition.

H.3 Model-Specific Frame-Manipulation Effects

Table 11 reports model-specific frame regressions for the four-model H3 subset. Each model-specific regression estimates Δ as a function of frame indicators and profile fixed effects, using the neutral frame as the reference condition. The pooled column corresponds to the main H3 specification reported in Table 5, which pools the same four models and includes model and profile fixed effects.

Table 11: Model-specific frame-manipulation effects

Frame contrast	GPT-4.1	Llama 3.1 70B	Qwen3.6-72B	Mistral Large	Pooled
Autonomy vs. neutral	3.07***	2.33***	1.85*	3.22***	2.62***
Collective obligation vs. neutral	-8.02***	-16.19***	-10.96***	-7.00***	-10.54***
Equity/access vs. neutral	5.37***	-0.11	12.89***	7.00***	6.29***

Notes: Each model-specific column reports a regression of Δ on frame indicators and profile fixed effects for the indicated model. The pooled column reports the main H3 specification for the four-model H3 subset with model and profile fixed effects. The neutral frame is the omitted reference category. Positive coefficients indicate larger simulated burden effects relative to neutral; negative coefficients indicate smaller simulated burden effects. Standard errors and 95% confidence intervals are reported in Table 12. * $p < 0.05$, *** $p < 0.001$.

Table 12 reports the corresponding standard errors and confidence intervals. The model-specific results show both robustness and heterogeneity within the H3 subset. The collective-obligation frame reduces simulated burden effects in all four models. The autonomy frame is positive in all four models. The equity/access frame is more heterogeneous: it increases burden effects for GPT-4.1, Qwen3.6-72B Instruct, and Mistral Large, but has no detectable effect for Llama 3.1 70B Instruct. Mistral Large is especially informative because it has the lowest PVC score and the smallest baseline burden effect in the expanded fifteen-model sample, yet all three frame manipulations shift its simulated burden effects in the expected directions. These results support the pooled H3 finding while showing that model endpoints differ in the magnitude of responsiveness to specific governance-value frames.

I Additional Robustness Checks

This appendix reports additional robustness checks for the H1 orientation-effect analysis. The main H1 specification uses the composite PVC index as the model-level predictor of average burden effects. Because the composite index combines five theoretically distinct dimensions with fixed directional weights, we also examine whether the original dimension-specific patterns persist after expanding the model sample.

Table 12: Model-specific frame-manipulation effects with standard errors and confidence intervals

Model	Frame contrast	Coefficient	SE	95% CI
GPT-4.1	Autonomy vs. neutral	3.07***	0.40	[2.29, 3.86]
GPT-4.1	Collective obligation vs. neutral	-8.02***	0.40	[-8.81, -7.24]
GPT-4.1	Equity/access vs. neutral	5.37***	0.40	[4.59, 6.16]
Llama 3.1 70B Instruct	Autonomy vs. neutral	2.33***	0.65	[1.05, 3.62]
Llama 3.1 70B Instruct	Collective obligation vs. neutral	-16.19***	0.65	[-17.47, -14.90]
Llama 3.1 70B Instruct	Equity/access vs. neutral	-0.11	0.65	[-1.39, 1.17]
Qwen3.6-72B Instruct	Autonomy vs. neutral	1.85*	0.77	[0.33, 3.37]
Qwen3.6-72B Instruct	Collective obligation vs. neutral	-10.96***	0.77	[-12.48, -9.45]
Qwen3.6-72B Instruct	Equity/access vs. neutral	12.89***	0.77	[11.37, 14.41]
Mistral Large	Autonomy vs. neutral	3.22***	0.44	[2.37, 4.08]
Mistral Large	Collective obligation vs. neutral	-7.00***	0.44	[-7.85, -6.15]
Mistral Large	Equity/access vs. neutral	7.00***	0.44	[6.15, 7.85]
Pooled	Autonomy vs. neutral	2.62***	0.34	[1.95, 3.29]
Pooled	Collective obligation vs. neutral	-10.54***	0.34	[-11.22, -9.87]
Pooled	Equity/access vs. neutral	6.29***	0.34	[5.61, 6.96]

Notes: Each row reports a frame contrast relative to the neutral frame. Model-specific regressions include profile fixed effects. The pooled regression includes model and profile fixed effects. The dependent variable is $\Delta = W^{low} - W^{high}$. Positive coefficients indicate larger simulated burden effects relative to neutral. * $p < 0.05$, *** $p < 0.001$.

I.1 Expanded-Sample H1 Robustness

Table 13 summarizes the key H1 robustness results for the original nine-model sample and the expanded fifteen-model sample. The composite PVC index does not predict model-level burden effects in either sample. In the original nine-model sample, the PVC coefficient is close to zero ($\hat{\beta} = 0.14$, $p = 0.956$, $R^2 = 0.0005$). After expanding to fifteen endpoints, the PVC coefficient remains weak and statistically indistinguishable from a null effect at conventional power ($\hat{\beta} = 1.51$, $R^2 = 0.043$).

The expanded sample also changes the interpretation of the dimension-specific collective obligation result. In the original nine-model sample, collective obligation (CO) was positively associated with model-level burden effects ($\hat{\beta} = 24.22$, $p = 0.009$, $R^2 = 0.652$). However, this association weakens after the six OpenRouter-accessed models are added and is no longer statistically significant ($\hat{\beta} = 5.68$, $p = 0.124$). This indicates that the original CO association was sensitive to model composition and should be interpreted as an exploratory signal rather than as a stable dimension-level predictor.

The expanded-sample measurement diagnostics also suggest caution in interpreting the composite PVC formula. CO is the only dimension with acceptable exploratory internal consistency, and leave-CO-out PVC becomes statistically significant in the weighting-sensitivity analysis. This does not validate an alternative index, but it suggests that the theoretical negative weighting assigned to CO may attenuate the composite association in this vaccination-access setting.

Table 13: H1 robustness across original and expanded model samples

Predictor and sample	Coefficient	p	R^2	Interpretation
PVC composite, $N = 9$	0.14	0.956	0.0005	Null
PVC composite, $N = 15$	1.51	0.456	0.043	Null
Collective obligation, $N = 9$	24.22	0.009	0.652	Initial signal
Collective obligation, $N = 15$	5.68	0.124	–	Not stable
Leave-CO-out PVC, $N = 15$	4.27	0.047	0.270	Exploratory sensitivity

Notes: Each row reports a separate bivariate model-level regression of $\bar{\Delta}_m$ on the listed predictor or PVC variant. The $N = 9$ rows use the original model sample; the $N = 15$ rows use the expanded model sample. Higher $\bar{\Delta}_m$ indicates a larger predicted reduction in willingness under high administrative burden. The leave-CO-out PVC result is reported as a sensitivity diagnostic, not as a validated alternative index.

I.2 Leave-One-Model-Out Diagnostics

Leave-one-model-out diagnostics further support the main H1 conclusion. The composite PVC coefficient is not a stable predictor of model-level burden effects. In the expanded sample, the bivariate PVC association remains weak and statistically indistinguishable from zero. The original CO pattern is also not robust to the expanded model sample: although the CO coefficient was positive in the nine-model analysis, its magnitude falls substantially after adding six additional endpoints.

These diagnostics reinforce the interpretation that H1 does not provide evidence for a stable aggregate orientation–effect relationship. The PVC index is useful as a theoretically

motivated diagnostic construct, but the expanded analysis does not support treating it as a validated scalar predictor of simulated burden-effect magnitude.

I.3 Power and Minimum Detectable Effect Diagnostics

The original nine-model H1 analysis was underpowered for small or moderate model-level associations. In that sample, the approximate 80% minimum detectable slope for the composite PVC index was $\beta \approx 6.8$. The observed PVC coefficient was far smaller than this threshold. Expanding the model sample to fifteen endpoints improves coverage but does not eliminate the basic limitation that H1 relies on model-level variation.

For this reason, the H1 estimates should be interpreted as descriptive diagnostics rather than high-powered population estimates. The expanded results are most informative as evidence against a large linear association between the composite PVC index and simulated burden effects in this model sample. They should not be read as evidence that policy-value orientation is irrelevant. Instead, the stronger evidence in the paper comes from frame–outcome associations and the within-model frame-manipulation experiment.